Review Poisoning: Understanding the Effect of LLM Generated Fake Reviews on Platform Chatbots

Short Paper

Introduction

Product reviews are increasingly read by two audiences: shoppers and platform shopping assistants. When an assistant uses reviews to answer a question such as whether a charger overheats, a fabricated first-person account can influence both the review page and the answer. Its effect may extend beyond the average star rating: the assistant could turn a seller-planted claim into concise advice. We examine this potential pathway, which we call review poisoning: deceptive reviews enter the evidence available to an LLM assistant and alter its product-specific answers. The manipulated material is ordinary review content, not an instruction addressed to the model.

Research on review fraud establishes the stakes but usually stops before the conversational answer. Sellers purchase fake reviews to raise ratings and sales, and such manipulation is concentrated among lower-quality products (He, Hollenbeck, & Proserpio, 2022). Even a limited injection of fake reviews can change a business's visibility in review-based rankings (Lappas et al., 2016). Platforms also select a small set of prominent reviews to help consumers process otherwise overwhelming information, making the composition of that set consequential (Jabr & Rahman, 2022). These studies show how manipulated reviews can affect marketplace signals; they do not establish whether a shopping assistant repeats a specific fabricated claim as fact.

Detection research addresses a different point in this chain. Using directly observed fake-review purchases, He, Hollenbeck, Overgoor, et al. (2022) show that products buying reviews cluster in product–reviewer networks because they rely on shared reviewers. Such structural signals can be harder to manipulate than review text, so the availability of LLM-generated prose does not, by itself, make existing detection methods obsolete. Recent work develops methods for identifying AI-generated deceptive reviews (Chernyaeva et al., 2025). Yet identifying suspicious reviews and ensuring that an assistant does not use their claims are distinct tasks. Indeed, fraudulent reviews can propagate into collaborative-filtering recommendations even after the reviews are removed (Adamopoulos, 2024). Whether a comparable spillover occurs in LLM-generated shopping answers remains an open empirical question.

The assistant adds a selection and synthesis step to the review system. It may retrieve a subset of reviews, compress conflicting accounts, and express a product conclusion without conveying the provenance or uncertainty of the underlying claims. A false assertion about battery life, safety, or compatibility could thus appear as corroborated consumer experience and affect a comparison between products. This is a plausible mechanism, not an assumed failure: an assistant might instead discount repetition, cross-check product specifications, or acknowledge conflicting evidence. The key issue is when a fabricated claim survives the journey from review text to answer and recommendation.

We investigate this issue with controlled review sets containing genuine reviews and varying proportions of LLM-generated deceptive reviews, while keeping independently verified product attributes available as ground truth. We ask:

RQ1: How does the proportion of LLM-generated deceptive reviews affect a shopping assistant's accuracy and uptake of fabricated product claims?

RQ2: Under what review and query conditions does the assistant repeat a false claim or recommend a product that is inferior on a verified attribute?

By tracing review contamination into the assistant's answer, the study extends research on review fraud from detecting or filtering deceptive content to assessing the reliability of the advice generated from it. It also identifies the conditions under which review provenance and corroboration matter for platform assistants.

Methodology

Research design and data. We propose a review-classification stage followed by two studies of shopping-assistant responses: an observational study of Amazon's Alexa for Shopping (formerly Rufus) and a controlled experiment with a review-grounded LLM assistant. Our historical corpus is McAuley Lab's Amazon Reviews 2023 dataset, produced at the University of California San Diego. It contains review text, ratings, product identifiers, timestamps, verified-purchase indicators, and product metadata, with interactions through September 2023 (McAuley Lab, 2024). We will select product categories with checkable attributes and active listings, link variants using ASINs and parent ASINs, and collect additional publicly visible reviews from the selected listings through 2026. We will document the collection dates, coverage, and any differences between the historical corpus and the current review pages.

Labeled review corpus. The training set will distinguish four sources: (1) authentic human reviews with documented provenance; (2) those authentic reviews copyedited by an LLM solely for grammar and clarity, with their factual content and valence preserved; (3) human-authored fabricated reviews collected through a controlled deceptive-writing task or independently validated corpus; and (4) fabricated reviews written by LLMs (Figure 1). We will generate the fourth group with zero-shot, one-shot, and few-shot prompts and fine-tuned generators. Both proprietary and locally run open-weight models, including Llama-family models, will be represented. We will vary product category, review valence, length, and the fabricated attribute while matching these features across groups. Copyedited genuine reviews remain authentic: AI assistance in wording is not itself evidence of deception. Likewise, a verified-purchase tag alone will not be treated as conclusive proof that a historical review is genuine.

Classifier development and application. We will train a text classifier to estimate review provenance and deception separately, comparing it with simpler lexical baselines. Training, validation, and test sets will be separated by product and source review to prevent near-duplicate leakage. Additional tests will hold out model families, prompting strategies, product categories, and later reviews to assess generalization beyond the generators seen during training. We will report class-specific precision and recall, macro-F1, calibration, and errors on a manually assessed sample. The fitted model will assign probabilities, rather than unquestioned truth labels, to reviews in the historical corpus and the 2024–2026 supplement. Product-level estimates of the share of suspected human and LLM-generated fakes will incorporate classification uncertainty and be checked against manual annotations.

Live-platform observation. For products with sufficient current reviews and independently verifiable attributes, we will ask Alexa for Shopping standardized product-specific, attribute-specific, and comparative questions. We will record the ASIN, visible review composition, listing and specification information, question wording, date, locale, account conditions, answer, and any displayed sources. Based on classifier probabilities, products will be grouped by the estimated presence of authentic human, human-fabricated, and LLM-fabricated reviews, including mixtures of all three. We will compare whether the assistant repeats claims appearing in suspected deceptive reviews, qualifies conflicting evidence, or recommends a product that is weaker on a verified attribute (Figure 1). Since Amazon controls which sources its assistant retrieves, and its answers can also draw on listings, Q&A, and web material (Amazon Staff, 2026), these comparisons will be interpreted as associations rather than experimental effects of particular reviews.

Controlled sensitivity experiment. We will build a separate retrieval-based shopping assistant whose review collection, retrieval settings, model version, and prompt can be fixed. For each product, the same genuine-review pool will be assigned to four conditions: genuine human reviews only; genuine plus human-fabricated reviews; genuine plus LLM-fabricated reviews; and genuine plus both types of fabricated reviews. We will vary the fabricated-review share across pre-specified levels while keeping the total number of reviews comparable, and will manipulate repetition of a false product claim and whether the question directly asks about that attribute. The fabricated claims will conflict with independently verified product information retained for evaluation. Across repeated runs and model families, we will log both retrieved passages and final answers to locate where the false claim enters the response. No fabricated reviews will be posted to Amazon.

Outcomes and analysis. The primary outcomes are false-claim uptake (asserting a fabricated attribute as fact) and recommendation error relative to the verified product attribute. Secondary outcomes include explicit uncertainty, contradiction handling, and attribution to review evidence. Two coders blinded to review condition will assess answers, with disagreements adjudicated. We will estimate how these outcomes vary with fake-review

share, review type, claim repetition, and query specificity, accounting for repeated observations within products and prompts. We will report the observational Alexa for Shopping patterns separately from the causal estimates of the controlled experiment.

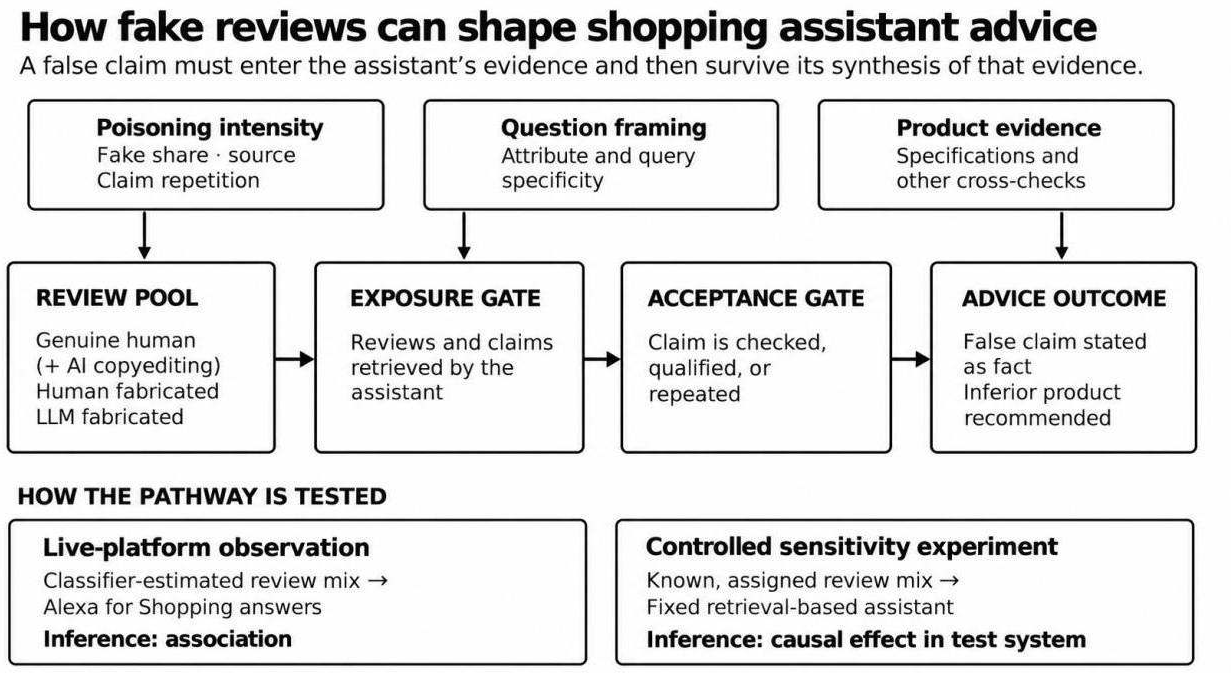


Figure 1*. Understanding pathways of review poisoning*

Conclusion

This paper proposes to examine whether fabricated reviews influence the platform chatbot's product advice generated from them. By combining review classification, observation of Alexa for Shopping, controlled retrieval experiment and a sensitivity analysis, we will distinguish the prevalence of suspicious content from the effect of known false claims on an assistant's answer. The experimental comparison will test whether fake-review share, authorship, claim repetition, and query specificity change false-claim uptake and product recommendations.

The proposed design has clear limits. Text classifiers cannot establish deception in unverified marketplace reviews, and Amazon's source selection is opaque; live-platform patterns will therefore be interpreted descriptively. Controlled manipulation can identify effects within the research assistant, but those effects cannot be assumed to transfer unchanged to Amazon. The study will identify whether and when review poisoning turns consumer testimony into apparently authoritative advice, including conditions in which assistants resist such contamination. Its findings can inform how shopping assistants assess review provenance and resolve conflicting product claims.

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